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# The Investigation of Supersonic Commercial Transport Performance Issues Using Multivariate Optimization

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## ABSTRACT

This paper describes the ASTRO multivariate optimisation code, comprising Supersonic Commercial Transport (SCT) conceptual design synthesis routines linked to a constrained optimisation method. The code currently optimises up to 33 independent variables, describing the aircraft and its operation, while applying up to 29 constraints. Aircraft take-off mass or operating costs can be minimised. The modeling used in the synthesis is described and two studies using ASTRO are presented, one showing the sensitivity of an SCT to changes in airfield noise requirements beyond current Stage 3 levels, and the other exploring the influence of subsonic leg length on the optimum configuration.

## INTRODUCTION

Any second generation Supersonic Commercial Transport (SCT) aircraft which intends to succeed in commercial service will have to meet a particularly challenging combination of performance goals to be viable. This will demand a significantly more integrated design than is needed for subsonic commercial aircraft, and require the identification of the best combination of configuration and technologies across the range of design disciplines. In developing an SCT concept it is therefore important to have tools that can find the best combination of design and technology parameters to meet a particular set of requirements. Constrained multivariate optimisation (MVO) is a process central to many such tools.

The Defence Evaluation & Research Agency (DERA) has considerable experience in the application of MVO to aircraft conceptual design for the purposes of assessing concepts, technologies and performance, and for sensitivity studies<sup>1-3</sup>. This experience is based on the use of RQPMIN, the DERA-developed constrained non-linear optimisation method, in conjunction with design synthesis codes for various classes of aircraft. British

Aerospace (BAe) has substantial experience of supersonic transport design from its Concorde involvement and has conceptual design methods which have been developed and updated during follow-on studies for any next generation SCT. DERA and BAe have collaborated on the development of an SCT design synthesis, with BAe methods being incorporated within DERA methodology, to produce the Advanced Supersonic Transport Optimisation (ASTRO) MVO code for joint studies and further development.

In developing ASTRO, priority was given to modeling those aspects of performance that were considered critical to aircraft viability. This naturally included cruise and low speed aerodynamic performance, but also airfield noise, which is not normally considered at the conceptual design stage, and operating costs which are important to economic viability.

The two short studies presented in this paper are preliminary, and exercise key parts of ASTRO's modeling. The first uses the jet noise estimation method to drive the optimisation, first to a Stage 3 (noise regulation) solution and then beyond to look at the consequences for SCT design of future reductions in noise limits. The second study looks at the influence of subsonic cruise requirements on the design of a supersonic transport.

## MULTIVARIATE OPTIMISATION WITH RQPMIN

ASTRO comprises an SCT design synthesis linked to the RQPMIN optimisation code. The RQPMIN optimiser finds the values of a set of  $n$  independent variables which minimise the value of a function (the objective function) of those variables, while simultaneously satisfying a set of  $m$  constraints. More precisely, if  $\mathbf{x}$  is the  $n$  independent variables  $x_1, x_2, \dots, x_n$  then RQPMIN minimises :

$f(\mathbf{x})$  the objective function

such that for  $i = 1, 2, \dots, n$  :

$$x_i^L \leq x_i \leq x_i^U \quad \text{independent variable bounds}$$

and for  $j = 1, 2, \dots, m$  :

$$c_j(\mathbf{x}) = 0 \quad \text{equality constraint}$$

or

$$c_j(\mathbf{x}) \leq 0 \quad \text{inequality constraint}$$

or

$$a_j \leq c_j(\mathbf{x}) \leq b_j \quad \text{double inequality constraint}$$

RQPMIN requires an input file containing starting values for the independent variables ( $\mathbf{x}_0$ ), their lower and upper bounds ( $x_i^L, x_i^U$ ), the constraint types (equality, inequality or double inequality) and the bounds of any double inequality constraints ( $a_j, b_j$ ). RQPMIN obtains values of the objective and constraint functions ( $f(\mathbf{x}), c_j(\mathbf{x})$ ) from the application code called as a subroutine, which in ASTRO is the SCT design synthesis. For a given optimisation run individual independent variables can either be fixed at their starting value or made available to RQPMIN to vary between their specified bounds. Individual constraints can be switched on or off as required.

The sets of independent variables and constraint functions currently used by ASTRO are shown in tables 1 and 2 respectively. Three alternative objective functions are shown in table 2, only one of which is used at a time.

## THE ASTRO DESIGN SYNTHESIS

The design synthesis performs all computations required to calculate the values of the objective and constraint functions that correspond to a set of independent variable values passed to it by RQPMIN. RQPMIN typically calls up the design synthesis routines several hundred times before converging to a solution, and often makes several thousand calls before doing so. Since multiple runs are normally required to explore a problem fully, there is a requirement to limit the run time, and hence the complexity, of the synthesis.

The ASTRO synthesis is highly modular in structure with separate top level subroutines for individual disciplines, such as geometry synthesis, mass estimation, mission modeling, and low-speed performance modeling. Below these are layers of more specific functional subroutines, of which some such as the drag estimation routines contain several alternative methods which can be selected via input variables.

**AIRCRAFT GEOMETRY** - This has been kept as simple as possible while still representing SCT configurations of interest sufficiently well to allow useful MVO studies to be performed. Figure 1 shows the typical configuration used by ASTRO, which currently does not include a tailplane or canard.

| Independent Variable  | Description                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------|
| 1 <sup>†</sup>        | Engine thrust scale factor,                                                                      |
| 2 <sup>†</sup>        | Fuselage length (m),                                                                             |
| 3 <sup>†</sup>        | Gross wing planform area (m <sup>2</sup> ),                                                      |
| 4 <sup>†</sup>        | Gross wing aspect ratio,                                                                         |
| 5                     | Spanwise location of leading edge kink as a proportion of the gross wing semi-span,              |
| 6                     | Inboard wing leading edge sweep angle (deg.),                                                    |
| 7                     | Outboard wing leading edge sweep angle (deg.),                                                   |
| 8 <sup>†</sup>        | Wing trailing edge sweep angle (deg.),                                                           |
| 9 <sup>†</sup>        | Wing root thickness/chord ratio,                                                                 |
| 10 <sup>†</sup>       | Wing 'intermediate' thickness/chord ratio,                                                       |
| 11 <sup>†</sup>       | Wing tip thickness/chord ratio,                                                                  |
| 12                    | Distance of gross wing apex aft of the fuselage nose, as a proportion of the fuselage length,    |
| 13                    | Net fin planform area (m <sup>2</sup> ),                                                         |
| 14                    | Distance of engine nacelle exit planes aft of the wing trailing edge (m),                        |
| 15                    | Spanwise location of inboard engine nacelle centreline, as proportion of gross wing semi-span,   |
| 16                    | Spanwise location of outboard engine nacelle centreline, as proportion of gross wing semi-span,  |
| 17 <sup>†</sup>       | Spanwise extent of wing fuel tanks, as a proportion of gross wing semi-span,                     |
| 18 <sup>†</sup>       | 'Intermediate' thickness/chord ratio spanwise location, as a proportion of gross wing semi-span, |
| 19 <sup>†</sup>       | Lift-off speed (m/s),                                                                            |
| 20 <sup>†</sup>       | Throttle cutback height for noise abatement (m),                                                 |
| 21 <sup>†</sup>       | Proportion of maximum fuel capacity carried for mission 1,                                       |
| 22 <sup>†</sup>       | Proportion of maximum fuel capacity carried for mission 2,                                       |
| 23 <sup>†</sup>       | Aircraft angle of attack on approach (deg.),                                                     |
| 25+5*I+J <sup>†</sup> | Start altitude of leg J of mission I (m),<br>{I=1,2 and J=1,2,3}.                                |

<sup>†</sup> Optimised in the presented studies.

Table 1: ASTRO Independent Variables

**The fuselage** - This consists essentially of a cylindrical centre section with simple tapers of equal length at both ends. The cross-section for a given width and height can be anything between an ellipse and a rectangle, specified by a single 'squareness factor', with consequent changes to cross-sectional and wetted areas. If some local reduction in fuselage width is to be used for area-ruling, to reduce aircraft maximum cross-sectional area and hence wave drag, then a minimum

acceptable fuselage width can also be specified. This is assumed to be located longitudinally wherever total aircraft cross-sectional area is a maximum.

| Function                   | Type    | Description                                                                                                                       |
|----------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1 <sup>†</sup>             | E       | Fuel used minus fuel carried, for mission 1 (kg),                                                                                 |
| 2 <sup>†</sup>             | E       | Fuel used minus fuel carried, for mission 2 (kg),                                                                                 |
| 3 <sup>†</sup>             | DI      | Wing tip chord length (m),                                                                                                        |
| 4 <sup>†</sup>             | I       | Sweep of supersonic cruise Mach cone, plus clearance, minus wing inboard leading edge sweep (deg.),                               |
| 5 <sup>†</sup>             | I       | Wing outboard leading edge sweep, plus clearance, minus sweep of supersonic cruise Mach cone (deg.),                              |
| 6 <sup>†</sup>             | I       | Take-off distance minus maximum acceptable distance (m),                                                                          |
| 7 <sup>†</sup>             | I       | Approach speed minus maximum acceptable speed (m/s),                                                                              |
| 8 <sup>†</sup>             | I       | Minimum acceptable streamwise separation of engine nacelle front face and wing leading edge, minus minimum actual separation (m), |
| 9 <sup>†</sup>             | I       | Minimum acceptable spanwise separation of inboard engine nacelles, minus actual separation (m),                                   |
| 10 <sup>†</sup>            | I       | Minimum acceptable spanwise separation of inboard and outboard engine nacelles, minus actual separation (m),                      |
| 11 <sup>†</sup>            | I       | Minimum acceptable second segment climb gradient, minus actual gradient,                                                          |
| 12                         | DI or E | Spanwise separation of 'intermediate t/c' location and wing leading edge kink,                                                    |
| 13 <sup>†</sup>            | I       | Minimum lift-off speed minus actual lift-off speed (m/s),                                                                         |
| 14 <sup>†</sup>            | I       | Minus the horizontal distance from end-of-bunt to flyover point (m),                                                              |
| 15 <sup>†</sup>            | I       | Sideline noise level, minus the noise limit, minus 2 EPNdB (EPNdB),                                                               |
| 16 <sup>†</sup>            | I       | Flyover noise level, minus the noise limit, minus 2 EPNdB (EPNdB),                                                                |
| 17 <sup>†</sup>            | I       | Approach noise level, minus the noise limit, minus 2 EPNdB (EPNdB),                                                               |
| 18 <sup>†</sup>            | I       | Sum of the three noise limit exceedance / satisfaction margins (EPNdB),                                                           |
| 19 <sup>†</sup>            | I       | Sum of all noise limit exceedances, minus 3 EPNdB (EPNdB),                                                                        |
| 20+10*I<br>+J <sup>†</sup> | I       | Minimum acceptable climb capability at the start of leg J of mission I, minus actual capability (m/s), {I=1,2 and J=1,2,3},       |
| 71 <sup>†</sup>            | O       | Take-off mass (kg),                                                                                                               |
| 72 <sup>†</sup>            | O       | Direct operating cost (\$/trip),                                                                                                  |
| 73 <sup>†</sup>            | O       | Cash operating cost (\$/trip).                                                                                                    |

Function Types: E Equality constraint  
I Inequality constraint  
DI Double Inequality constraint  
O Objective function

<sup>†</sup> Used in the presented studies.

Table 2: ASTRO Constraint & Objective Functions

**The wing** - The wing planform has been kept as simple as possible with just two leading edge sweep angles and a single trailing edge sweep angle. Aerofoil shapes are defined either by using a combination of simple elliptic and power law expressions, fore and aft of maximum thickness respectively, or by using the expressions employed by Hutchison et al<sup>4</sup> which are based on the NACA 4-digit definition. The former method allows a trailing edge thickness to be specified and both methods include options for either round or sharp leading edges. Thickness/chord ratio is defined at three spanwise locations (root, tip and an 'intermediate' location) and interpolated between them. Leading and trailing edge high-lift devices of required sizes can also be simply specified.

**The engines and nacelles** - A set of data for a datum size engine is supplied as input. This includes size and mass data plus a set of performance data for a range of heights, Mach numbers and throttle settings. These data are scaled by functions of the engine thrust scale factor (an independent variable) to allow changes in engine size to be represented. Nacelle size is related to engine size via a set of parameters input by the user. The shape of the nacelle cross-section can be selected from four options, including a 'paired' Concorde style arrangement. Circular streamtubes are subtracted from the cross-sectional area when calculating the contribution of the nacelles to the aircraft maximum cross-sectional area for wave drag estimation. Nacelle location on the wing is controlled by independent variables and constraints.

**The fin** - Although fin area is an independent variable there is currently no modeling in ASTRO to drive its size. Current studies therefore fix this value, along with the other input variables which determine fin shape.

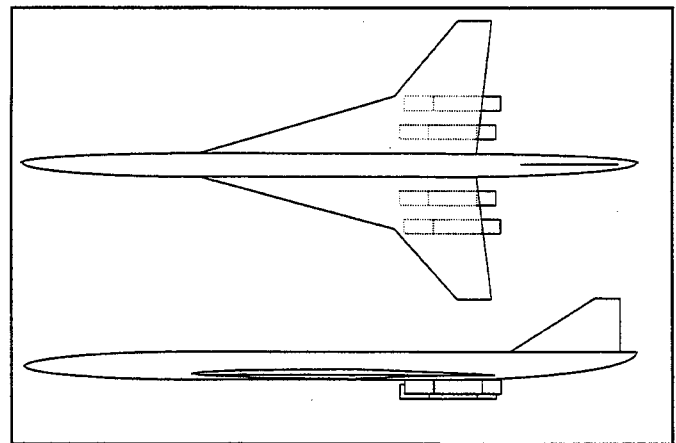


Figure 1: ASTRO SCT representation

**MASS ESTIMATION** - Most of the mass estimation in ASTRO is based on simple BAe expressions which were developed with Concorde experience and follow-on studies. A number have been updated recently to give improved agreement with more detailed methods for the

SCT configurations of current interest. For example the wing mass expression in ASTRO now matches data from a buckling panel theory method with an average error of 2% across a range of wing areas from 650 to 930m<sup>2</sup> and aspect ratios from 1.7 to 2.6. Factors have been incorporated into the mass estimates of most components to allow mass reduction technologies relative to a datum to be easily represented and adjusted.

**DRAG ESTIMATION** - The drag of the complete aircraft is divided into skin friction drag, wave drag and lift-dependent drag for estimation purposes. The engine performance data supplied as input incorporate spillage drag.

Skin friction drag - This is calculated using a DERA method which multiplies the wetted area of each major aircraft component (fuselage, inner and outer wing, nacelles, fin) by a mean turbulent skin friction coefficient of a flat plate of uniform roughness at a Reynolds number based on an appropriate component length. A component form factor is then applied. The resulting component drags are multiplied by individual technology factors which are supplied as inputs to allow the aerodynamic benefits of technologies such as laminar flow to be represented.

Wave drag - ASTRO currently contains two alternative wave drag methods selectable via an input variable:

A BAe method uses simple fuselage and wing geometric parameters to estimate the drag of the aircraft with no engines, and then applies an engine installation factor. This method currently only achieves acceptable accuracy for a limited range of configurations and requires further development.

A method based on the approach of Raymer<sup>5</sup> is also included, which gives a wave drag proportional to  $(S_{\max}/L_{\text{fus}})^2$  where  $S_{\max}$  is the maximum cross-sectional area of the aircraft and  $L_{\text{fus}}$  is the fuselage length. An efficiency factor is also applied which allows the method to be matched to measured data with a known level of area-ruling. Although the expression used is fairly simple the method requires a considerable amount of computation due to the need to search longitudinally for  $S_{\max}$ , which includes contributions from the fuselage, wing and usually the engine nacelles. This search requires accurate estimation of aircraft cross-sectional area at a range of longitudinal positions, which in turn requires careful modeling of features such as the wing-fuselage junction. The advantage of the method is that it controls maximum aircraft cross-sectional area, a good first-order wave drag reduction strategy.

Lift-dependent drag - At subsonic Mach numbers lift-dependent drag is interpolated from arrays of BAe aerodynamic data for SCT wings which currently cover aspect ratios from 1.7 to 2.6. These data cover both

subsonic cruise and low speed  $C_L$  ranges, and include drag changes due to the deflection of datum sized leading and trailing edge flaps. Simple scaling of the datum high-lift device data, based on flap area, is used to estimate the effects of alternative device sizes.

For lift-dependent drag in supersonic cruise a BAe expression based on wing planform geometry is used. This matches an array of results from a more sophisticated and computationally intensive linearised theory method.

**MISSION MODELING** - Mission definitions are input to ASTRO and constraints are used to ensure that the optimised aircraft can complete them. Currently a maximum of two separate missions can be used. Individual missions are defined as a series of cruise-climb legs of specified lengths and Mach numbers. The leg lengths are automatically adjusted by ASTRO to model the additional fuel used in changing height and speed between legs. The starting altitude of each leg is an independent variable, early experimentation showing that this freedom can result in considerable improvements in aircraft performance. If two missions are specified, and operating costs are to be used as the objective function, one of the missions must be nominated for the operating costs estimation.

**NOISE ESTIMATION** - ASTRO uses a BAe jet noise prediction method based on measured data for Concorde with Olympus engines which has been modified to simulate a more general powerplant without reheat. Adjustments to represent advances in jet pipe liners and the elimination of tones since Concorde have also been included. The method estimates the sideline, flyover and approach noise levels used for aircraft noise certification and requires engine thrust, jet velocity, jet pressure ratio and aircraft height data at each flight condition of interest. The three engine parameters are interpolated from the input engine data file and the method can only represent those features of SCT engine concepts that can be expressed via these parameters. Turbomachinery and airframe noise are not currently calculated although they could become important if jet noise is sufficiently reduced.

An integral part of the noise estimation process is the modeling of the take-off and approach flight paths, in order to obtain aircraft heights and throttle settings at the noise certification conditions. ASTRO models the flight paths shown in figure 2, including the noise abatement 'bunt' manoeuvre on take-off. This manoeuvre allows the engines to be throttled back at a height  $z_{\text{cutback}}$  (an independent variable), to a setting that can maintain a specified minimum climb gradient. This reduces the vertical distance for which the engines are at their maximum, take-off, setting. The lower the aircraft, the shallower its angle above the ground when viewed from the sideline distance of 450m, which increases the ground attenuation of noise measured at the sideline

distance. Hence a reduction in  $z_{\text{cutback}}$  reduces the sideline noise. However, for a fixed aircraft a reduction in  $z_{\text{cutback}}$  also causes a reduction in the flyover height, i.e. the height that the aircraft passes overhead the flyover noise measurement point. This increases flyover noise since this is not dependent on ground attenuation effects. Since  $z_{\text{cutback}}$  is an independent variable in ASTRO it can be used during optimisation as a means of trading sideline and flyover noise as required to aid satisfaction of the complete set of noise constraints.

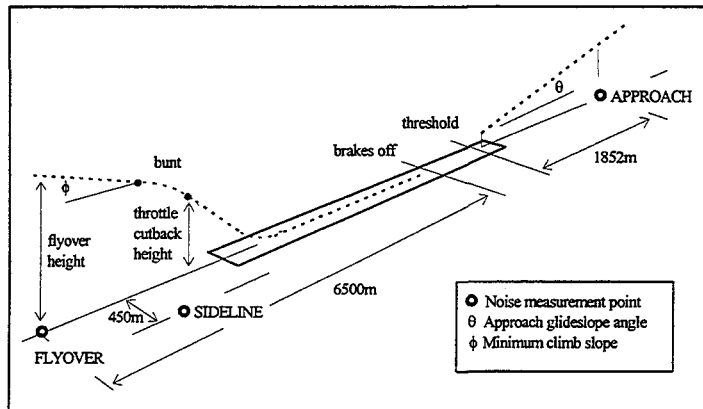


Figure 2: Airfield noise measurement geometry

## NOISE LIMIT SENSITIVITY STUDY

Any new SCT which intends to succeed Concorde will have to meet the same airfield noise regulations that apply to its subsonic contemporaries. Designing an SCT to meet current FAR36 Stage 3 noise limits is already a demanding task, but these limits may themselves be superseded by more stringent ones within the timescale of a new SCT. Hence the ability of SCT designs to meet post-Stage 3 noise regulations is an important issue.

ASTRO was used to perform a study which investigated the sensitivity of an SCT aircraft with Mid-Tandem Fan (MTF) engines to changes in airfield noise limits. A configuration which satisfies the Stage 3 noise limits was first optimised to provide a baseline aircraft. The noise limits were then reduced (i.e. made more severe) while retaining the same rules of application, the aircraft reoptimised, and the effects on the optimum configuration investigated.

**NOISE LIMIT APPLICATION** - The three main noise levels currently used for regulation are sideline, flyover and approach noise, which are measured at the three locations shown in figure 2. The current FAR36 Stage 3 limits at these locations are:

### Sideline:

103 EPNdB for  $m_{\text{TOMax}} \geq 882000$  lb (~400 tonnes),  
reducing by 2.56 EPNdB per halving of 882000 lb,  
down to 94 EPNdB for  $m_{\text{TOMax}} \leq 77200$  lb (~35 tonnes).

### Flyover (for aircraft with more than 3 engines):

106 EPNdB for  $m_{\text{TOMax}} \geq 850000$  lb (~386 tonnes),  
reducing by 4 EPNdB per halving of 850000 lb,  
down to 89 EPNdB for  $m_{\text{TOMax}} \leq 44673$  lb (~20 tonnes).

### Approach:

105 EPNdB for  $m_{\text{TOMax}} \geq 617300$  lb (~280 tonnes),  
reducing by 2.33 EPNdB per halving of 617300 lb,  
down to 98 EPNdB for  $m_{\text{TOMax}} \leq 77200$  lb (~35 tonnes).

where  $m_{\text{TOMax}}$  is the aircraft maximum take-off mass. It is important to note that these limits are not individual absolute limits, but are applied via the following Stage 3 rules:

1. Any individual limit may be exceeded by a margin of up to 2 EPNdB,
2. The sum of the three margins (+ve for limit exceedance, -ve for limit satisfaction) must be zero or less,
3. The sum of all exceedance (+ve only) margins may not exceed 3 EPNdB.

Hence it requires five constraints (15-19 in table 2) rather than three, to apply Stage 3 limits in ASTRO.

**STUDY ASSUMPTIONS** - To perform the study it was necessary to identify a representative set of performance requirements for an SCT, decide which design parameters should be fixed and which should be optimised, and what technology levels would be appropriate. The optimised aircraft was required to achieve the following performance:

- A range of 5200nm, all cruise at Mach 2.0, while carrying 253 passengers in 6 abreast seating,
- Maximum take-off distance of 3353m (11000ft),
- Maximum approach speed of 82.3m/s (160kts),
- Minimum 2nd segment climb gradient of 3%,
- Minimum climb capability in cruise of 2.54m/s (500ft/min),
- Minimum throttle cutback height of 150m,
- Meet Stage 3 (or stricter when specified) noise requirements.

The minimum throttle cutback height was set lower than the FAR36 Stage 3 value of 210m to give greater flexibility to meet challenging noise constraints. The independent variables that were optimised were:

Engine thrust scale factor,  
Wing planform area,  
Wing aspect ratio,  
Wing trailing edge sweep,  
Wing thickness/chord ratios,  
Fuel tank size,  
Lift-off speed,  
Throttle cutback height,  
Angle of attack on approach,  
Altitude at start of cruise.

The Mid-Tandem Fan (MTF) variable cycle engine was considered appropriate for the study since it is one of the more promising concepts identified to date<sup>6</sup>. It is able to limit airfield noise by changing its cycle at low speed to give reduced jet velocities. The engine used for the study was designed for a cruise speed of Mach 2.0 and had its nozzle exit jet velocity at take-off restricted to 400m/s to limit noise. A technology standard considered representative of the near future was used within the engine, including a maximum continuous compressor delivery temperature ( $T_3$ ) of 950K and a maximum continuous stator outlet temperature of 1750K (Olympus values are 850K and 1410K respectively).

The fuselage structure of the study SCT assumed a 20% mass reduction relative to Concorde due to improved manufacturing and Aluminium alloys, and a further 10% reduction due to the conservative use of composites. The wing structure assumed similar reductions through the use of Titanium and composites. A factor was applied to the fuselage mass of the study aircraft to represent an increased stiffness requirement relative to Concorde, due to increased fuselage slenderness.

The study assumed that only limited area-ruling of the fuselage would be used, and in particular that the passenger cabin width did not narrow to less than six abreast seating, except at the very front and back as part of the nose and tail tapers. Simple leading and trailing edge flaps, each of approximately 8% gross wing area, were assumed to be present. The leading edge flaps were used on take-off and in subsonic cruise for drag reduction, while the trailing edge flaps were only deployed during the approach to provide additional lift. Initial runs of ASTRO for the study aircraft revealed that in the absence of trailing edge devices significant increases in wing area, and hence aircraft mass and drag, were necessary to obtain sufficient lift on approach to achieve acceptably low approach speeds and noise levels. The pitching moments caused by high-lift device deployment, and the consequent trim and control power requirements, are not currently modeled by ASTRO. More detailed modeling of the low speed control requirements of SCTs by DERA and BAe has however validated ASTRO's low speed performance estimation.

**BASELINE STAGE 3 CONFIGURATION** - The optimised Stage 3 configuration obtained from ASTRO is summarised in table 3. This shows that the approach noise constraint is the only active noise constraint, approach noise being 2.0 EPNdB above its limit. Sideline and flyover noise are both well below their respective limits so throttle cutback height is not strongly driven in a particular direction and takes a fairly arbitrary value. The value of the cutback height happens to be above the Stage 3 minimum of 210m, despite the use of a 150m minimum for this study, which confirms the baseline configuration as a full Stage 3 solution.

| Parameter                                          | Optimised value                          |
|----------------------------------------------------|------------------------------------------|
| Fuselage length                                    | 89.0 m *                                 |
| Gross wing area                                    | 930 m <sup>2</sup>                       |
| Gross wing aspect ratio                            | 1.90                                     |
| Wing trailing edge sweep                           | 10.0 deg. U                              |
| Wing root t/c                                      | 0.031 L                                  |
| Wing intermediate t/c                              | 0.021 L                                  |
| Wing tip t/c                                       | 0.021 L                                  |
| Maximum thrust per engine (sea level, static)      | 314 kN                                   |
| Lift-off speed used                                | 108 m/s                                  |
| Minimum lift-off speed                             | 108 m/s                                  |
| Throttle cutback height                            | 235 m                                    |
| Flyover height                                     | 390 m                                    |
| Approach speed                                     | 82.0 m/s                                 |
| Approach angle-of attack                           | 9.96 deg.                                |
| Sideline noise: estimated limit (limit exceedance) | 100.1 EPNdB<br>102.4 EPNdB<br>-2.3 EPNdB |
| Flyover noise: estimated limit (limit exceedance)  | 102.8 EPNdB<br>105.3 EPNdB<br>-2.5 EPNdB |
| Approach noise: estimated limit (limit exceedance) | 107.0 EPNdB<br>105.0 EPNdB<br>+2.0 EPNdB |
| Height at start of supersonic cruise               | 15660m<br>(51370 ft)                     |
| Supersonic cruise L/D                              | 8.49                                     |
| Operating mass empty                               | 142600 kg                                |
| Payload mass                                       | 24000 kg *                               |
| Fuel mass                                          | 178600 kg                                |
| Take-off mass                                      | 343600 kg                                |

\* Fixed value, does not change during optimisation.

L Independent variable at its lower bound.

U Independent variable at its upper bound.

Table 3: Optimised Stage 3 SCT configuration

**NOISE LIMIT SENSITIVITIES** - ASTRO was then run as for the baseline case except that each of the three noise limits was reduced in turn, i.e. made more severe, while the remaining two were kept at the Stage 3 values. The rules governing the application of the limits were kept the same. The effect on the optimised configuration of changing an individual limit was thus obtained and these results are shown in figure 3 for all three limits. The figure also shows the effect of reducing the sideline and flyover limits simultaneously and equally. Note that figure 3 shows the lowest noise limits for which solutions could be found, the curves cannot be extrapolated to higher take-off masses since such configurations would not satisfy all the constraints.

It is striking how steeply the take-off mass increases once any available 'slack' between noise level and noise limit has been taken up by reducing the limit. This indicates that substantial mass penalties are associated with redesigning the aircraft hardware, in terms of the set of independent variables optimised, to counter active noise constraints. Of particular concern is the approach noise curve which climbs most steeply and which shows that no reduction in the approach noise limit can be accommodated without a mass increase, and that the maximum reduction for which a solution can be found is only -0.35 EPNdB. It should be noted that the Stage 3

approach noise limit (defined earlier) does not increase with take-off mass for the plotted mass range, whereas the flyover and sideline limits do up to 386 and 400 tonnes respectively.

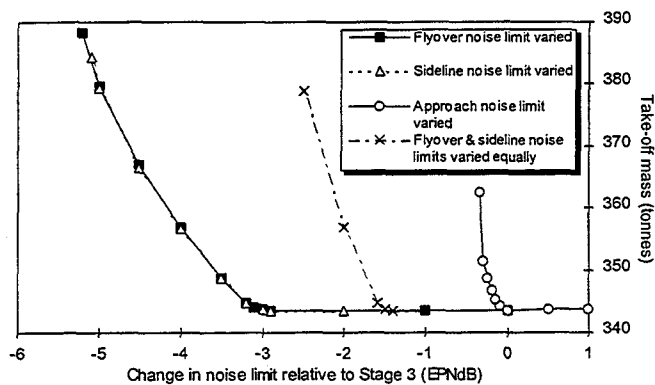


Figure 3: SCT+MTF sensitivity to changes in noise limits

**Approach noise limit sensitivity** - For a given glideslope angle the main parameter that affects ASTRO approach noise estimates is approach thrust. This is influenced by a number of parameters including aircraft mass, wing planform, high-lift device deployment, and approach speed. The baseline Stage 3 aircraft for the study had an approach speed very close to the maximum acceptable value of 160kts, so faster approaches to reduce thrust and hence noise could not be used. The aircraft mass was already being minimised, and high-lift device configuration was kept fixed during optimisation, so alterations to the wing planform were the main means of accommodating reductions in approach noise limits. Both wing area and aspect ratio were increased to achieve small reductions in approach noise, but the mass penalties of these changes made further noise reductions more difficult. Noise reduction by this method therefore results in rapidly increasing mass penalties followed by an absence of solutions, as demonstrated by the approach noise limit sensitivity curve of figure 3. It is purely coincidence that this curve starts to rise at 0 EPNdB on the x axis.

**Sideline and flyover noise limit sensitivity** - The sensitivity curves for sideline and for flyover noise limits in figure 3 are virtually identical, both accommodating noise limit reductions of -3.0 EPNdB before identical mass rises become necessary. Examination of the optimised configurations for the two curves confirmed that similar configurations were obtained for the same reduction in either noise limit, but with substantial differences in throttle cutback height. This indicates that sideline and flyover noise levels can be traded on an approximately one-for-one (EPNdB) basis, without a mass penalty, by simply varying throttle cutback height. This is confirmed by the curve for simultaneous reduction of sideline and flyover limits, which allows exactly half the noise limit reduction that can be accommodated if only one of the two limits is reduced.

Figures 4-6 show the values of key parameters and noise constraints corresponding to the flyover noise sensitivity curve of figure 3.

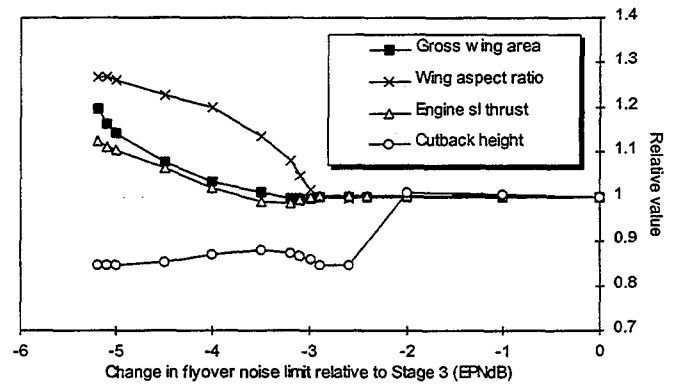


Figure 4: Variation of key independent variables with flyover noise limit

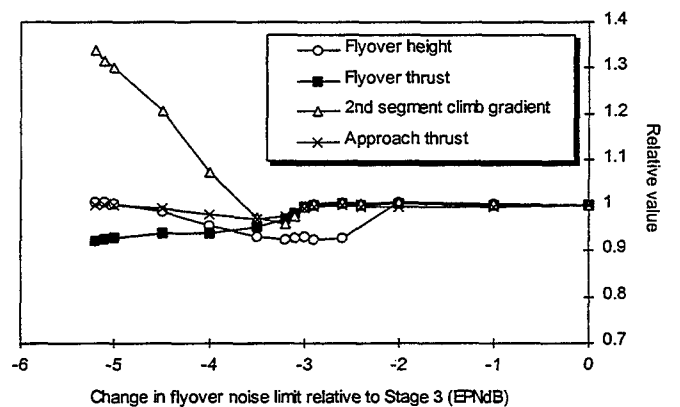


Figure 5: Variation of key parameters with flyover noise limit

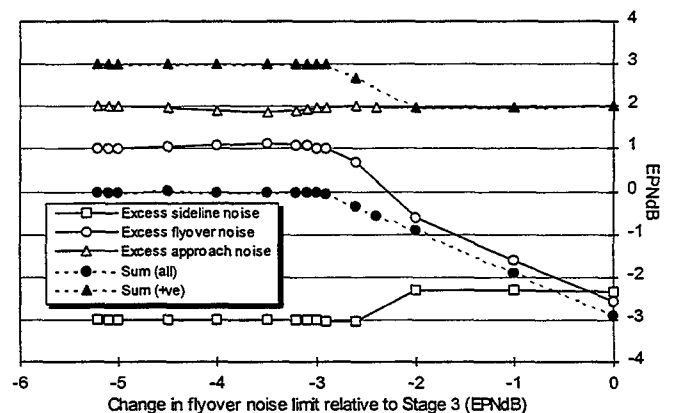


Figure 6: Variation of noise constraints with flyover noise limit

Figure 4 indicates that the initial rise in take-off mass from -3 to -3.3 EPNdB in figure 3 is due to a sharp rise in wing aspect ratio, the mass penalty of which is partly countered by a small reduction in engine size. The increased aspect ratio decreases drag, and hence thrust, at the flyover and approach conditions as confirmed in figure 5. The reduced flyover thrust, and hence noise, is not sufficient to fully counter the flyover noise limit reduction however, and the excess flyover noise rises above 1 EPNdB as shown in figure 6. The reduced

approach thrust reduces excess approach noise below 2 EPNdB to counter the increased excess flyover noise within the noise rules.

As the flyover noise limit was reduced below -3.3 EPNdB relative to Stage 3, increased aircraft mass required an increase in wing area to hold approach speed at its maximum acceptable value of 160kts. Engine thrust was also increased sufficiently to increase aircraft climb performance (which is determined by (thrust-drag)/weight), as shown in figure 5. The increased climb performance had the surprising result of allowing flyover height to increase as throttle cutback height fell. This was because the improved climb performance had reduced the horizontal distance taken by the initial acceleration and climb between lift-off and throttle cutback. Consequently, increased horizontal distance was available between cutback and the flyover noise measurement point in which to continue gaining height, and this more than countered the reduced cutback height. The increased flyover height, coupled with continued reductions in flyover thrust allowed flyover noise to be reduced in line with reductions in the noise limit, down to -5.3 EPNdB below the Stage 3 limit.

**FURTHER NOISE REDUCTION MEASURES** - Since the study aircraft had virtually no tolerance to reductions in the approach noise limit, measures that reduce approach noise were of particular interest. Approach noise is sensitive to approach speed, with increases in speed reducing noise, but the baseline aircraft already had an approach speed very close to the specified limiting value of 160kts. The latter was set on the basis of Concorde experience and it is considered unlikely that speeds significantly in excess of this would be acceptable in normal service. Runs of ASTRO with the approach speed limit raised to 200kts revealed that such a speed increase only accommodated an additional 1.0 EPNdB reduction of the approach noise limit.

Increasing the approach glideslope angle above the normal three degrees was considered as an alternative method of approach noise reduction. This places the aircraft higher above the noise measurement point, although the implications for safety, comfort and undercarriage mass of increased sink rates are concerns and compatibility with other air traffic could also be a problem.

A four degree glideslope was used to calculate a new set of noise limit sensitivities for approach and flyover noise, and figure 7 shows these superimposed on the three degree sensitivities. The above mentioned concerns regarding increased glideslope have not been addressed in deriving the new sensitivities. A simple inverse square reduction in approach sound pressure level with height above the approach noise measurement point was assumed, which gave a 2.2 EPNdB reduction in approach noise for a fixed aircraft at a fixed throttle setting. Comparison of the two 'A' curves in figure 7

reveals that the glideslope change reduces the approach noise of the optimised aircraft by 3.0 EPNdB, which comprises the 2.2 EPNdB due to increased height and a further 0.8 EPNdB due to reductions in approach thrust.

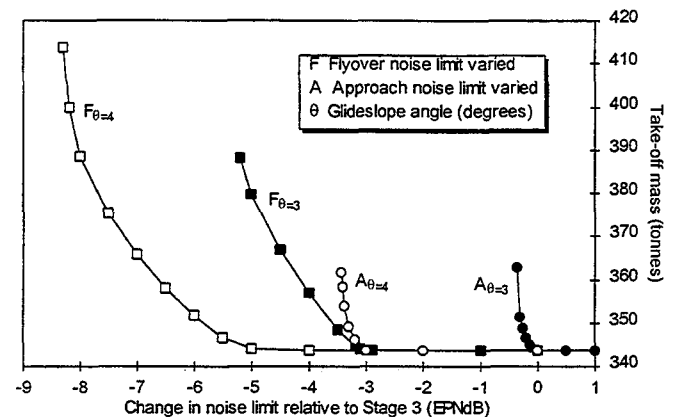


Figure 7: Effect of glideslope angle on noise limit sensitivities

A reduction in approach noise can either be used directly to accommodate reductions in approach noise limit, as shown by the horizontal shift between the two 'A' curves in figure 7, or it can be used more indirectly, through the noise limit application rules, to accommodate changes to the other noise limits. If approach noise is reduced but its limit is not, then excess approach noise (= noise minus noise limit) is reduced. This reduction can be used within the Stage 3 rules to counter increases in excess sideline or flyover noise, i.e. to accommodate reductions in their limits. The horizontal displacement between the two 'F' curves in figure 7 illustrates the reduction in flyover noise limit that can be accommodated by the glideslope angle change if the approach noise limit is not altered.

The presented results are valid for the stated set of study technology assumptions, including a fixed MTF engine cycle with a jet velocity of 400m/s at take-off. The sensitivity of the results to these assumptions will be investigated in follow-on work to identify which technologies are most effective at reducing airfield noise. Such technologies may include structural mass reduction, improved high-lift devices and different engine cycles, including reduced engine jet velocity.

## SUBSONIC LEG SENSITIVITY STUDY

The route network envisaged for a future SCT will principally involve long haul flights over water covering the important trans-Atlantic and trans-Pacific markets. However, there will be a need to fly some overland segments and these become more significant if routes such as Europe to the west coast of the USA or even Europe to the Far East are included in order to capture more of the world market.

Sonic boom is a problem for overland supersonic flight and aircraft designs to lower the boom's magnitude have reduced performance. Studies have suggested that it is not possible to produce a viable design with an

acceptable level of sonic boom. Hence, the current assumption is that supersonic flight will not be allowed over land apart from the possibility of using special 'corridors' across uninhabited areas.

Overland flight at high subsonic cruise speeds will affect the configuration design depending upon the extent of this overland segment. Hence ASTRO has been used to investigate the sensitivity of an optimum SCT aircraft design as the proportion of subsonic leg is varied.

**STUDY ASSUMPTIONS** - The assumptions for the study were the same as for the noise limit sensitivity study except that the total range was fixed at 5700nm. All configurations in this study were designed to meet Stage 3 noise limits. The objective function minimised during the optimisation was again set as take-off weight.

The configurations were optimised to meet the requirements of two missions, one having the subsonic leg before the supersonic cruise and the other having it after (i.e. an outbound subsonic leg and an inbound subsonic leg respectively). This used the two separate missions that can be modeled within ASTRO. All subsonic cruise legs were assumed to be at Mach 0.95 with supersonic cruise remaining at Mach 2.0.

An SCT will always travel a small distance at subsonic speeds during the climb and descent phases but an all supersonic mission in this study was said to have a zero subsonic range. Both the supersonic cruise and the subsonic cruise were assumed to be cruise climb legs.

**STUDY RESULTS** - The subsonic leg length was varied from 0% to 90% of the 5700nm total range. In each case, the configuration was re-optimised whilst meeting all constraints. Figure 8 shows how the main wing design parameters, aspect ratio and gross wing area vary. The wing area is seen to slowly decrease although the changes are relatively insignificant. In contrast, wing aspect ratio, a parameter which will help the subsonic cruise aerodynamic performance, increases as the subsonic leg becomes more significant. The lack of increase in aspect ratio for subsonic legs above 60% is due to the upper bound on the aspect ratio variable being reached. It would be expected to continue rising if this limit had not been set within the input files. The reason for setting the limit was that some of the methods within ASTRO had not been validated above this limit or drag data were not available.

The changes to the wing geometry have affected the aerodynamic performance in terms of lift/drag ratio as given in figure 9. The subsonic cruise performance is seen to increase with subsonic leg length and the opposite occurs with the supersonic performance. Most of this is because of the change in aspect ratio although there are additional variations due to changes in cruise altitude.

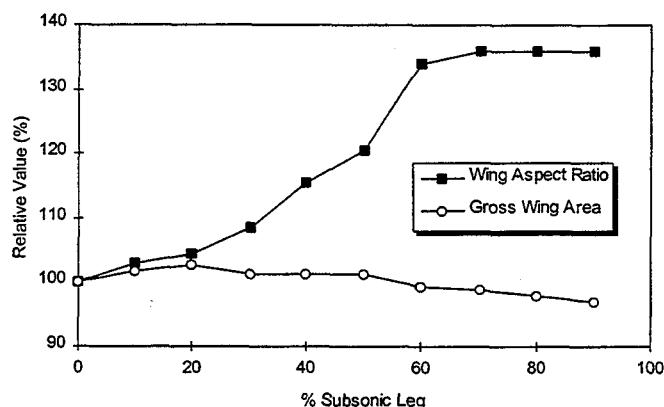


Figure 8: Variation of key wing variables with subsonic leg

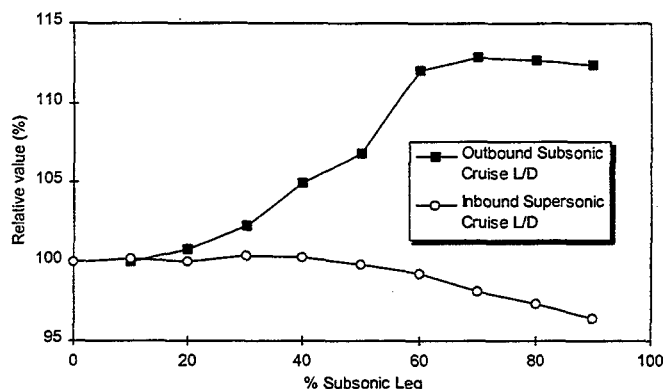


Figure 9: Variation of lift/drag

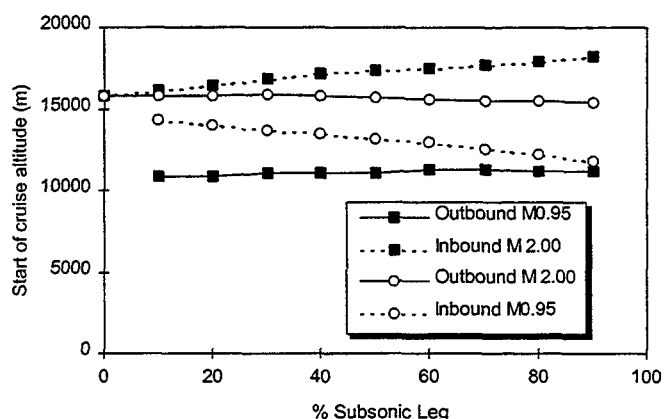


Figure 10: Variation of start of cruise altitude

The start-of-cruise altitudes for each leg have been set as independent variables which ASTRO will vary to minimise the objective function, in this case take-off weight. They will depend on start-of-cruise weight, the variation of L/D with  $C_L$  and engine performance. For example, assuming an increasing outbound subsonic cruise leg, the subsequent supersonic cruise will start at lower and lower weights. It was found that the optimum cruise altitude increased, to approximately keep a constant  $C_L$ . However, with the subsonic cruise as the second cruise leg, the  $C_L$  was allowed to rise to take full advantage of the better induced drag from the increasing aspect ratio. The start altitude of the inbound subsonic leg was thus seen to decrease. Figure 10 shows the

variation in altitudes resulting from the optimisation process.

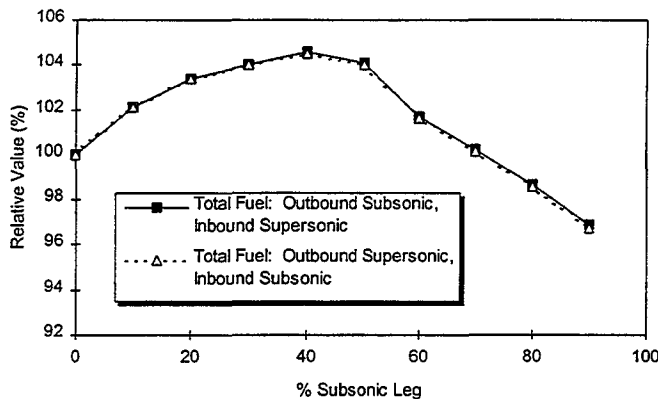


Figure 11: Variation of total fuel required for mission

The total fuel required for the mission includes the block fuel and reserves. Its variation with increasing proportion of subsonic leg is given in figure 11 for both missions, an outbound subsonic cruise and an inbound subsonic cruise. It can be seen that the position of the subsonic cruise segment did not make much difference. For interest, this was not true if the subsonic segment was placed in the middle of the supersonic cruise. This would require two supersonic climb/acceleration phases which penalise the mission heavily and for the assumed baseline aircraft, led to no possible solutions.

From figure 11, the amount of fuel is seen to increase as the subsonic leg proportion increases up to 40% of total range. Above this, the total fuel required decreases again such that at 70% subsonic cruise, the fuel used becomes less than the all supersonic cruise case.

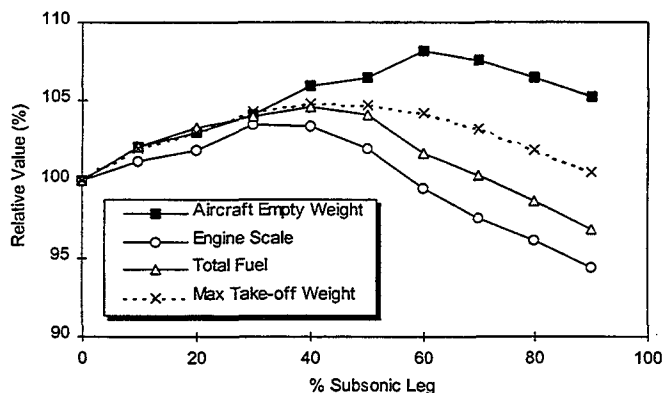


Figure 12: Variation of key parameters with subsonic leg

Figure 12 shows key parameters of the optimum aircraft configurations and their variation with increasing subsonic leg. With the wing aspect ratio increasing, a weight penalty is introduced which is only slightly offset by the decreasing wing area. Thus, the aircraft empty weight tends to increase which is made worse by the snowball effect of increasing take-off weight, combined with increasing engine scale. The engine scale is being driven by the take-off requirements. Thus up to 40% subsonic leg, the key parameters empty weight, engine

scale, fuel used and take-off weight are interacting in such a way that they all increase.

As the subsonic leg increases further, the increasing aspect ratio improves the subsonic performance to a point where the specific range is better during subsonic cruise than supersonic cruise. It also starts to help the low speed performance thus reducing the engine scale. These effects then dominate, reducing mission fuel and hence take-off weight.

When the aspect ratio reaches its upper bound at about 60% subsonic leg, the rate of improvement of the parameters is reduced but continues due to the better subsonic specific range.

For the given optimum configurations, above 50% subsonic cruise leg the ability to achieve an all-supersonic range of 5700nm is lost. This is unlikely to be acceptable for a true SCT configuration and hence some runs were made with the all-supersonic cruise being one of the required missions. Above 50% subsonic leg, the aspect ratio was then found not to increase any further and hence engine scale and take-off weight varied little for the new configurations.

**USE OF DIFFERENT OBJECTIVE FUNCTIONS** - As already mentioned, ASTRO has the ability to calculate operating costs and use them as alternative objective functions. Cash operating costs (COC) includes the flying costs and the maintenance costs of the aircraft. The direct operating costs (DOC) are the cash operating costs plus the financial costs to an airline of buying and owning the aircraft. The DOCs are believed to be the costs that can be affected by the aircraft manufacturer.

Figure 13 shows the variation in COC and DOC per trip with subsonic leg length for the previous study. Although there is some effect from the variation in take-off weight and fuel used, as seen by the change in slope of the COC line, the changes in costs are dominated by the variation in flight time or block time.

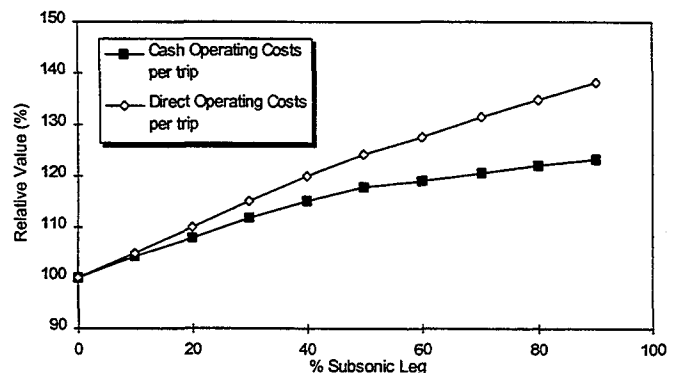


Figure 13: Variation of costs with subsonic leg

As the subsonic leg increases in length, the block time rapidly increases from 5.58 hours (0% subsonic) to 10.37 hours (90% subsonic). This increases crew costs but

more importantly reduces the utilisation of the aircraft in terms of number of trips per year. Thus, the maintenance costs per trip increase the COC and then the DOC are increased even further by the financial costs being spread over fewer trips per year.

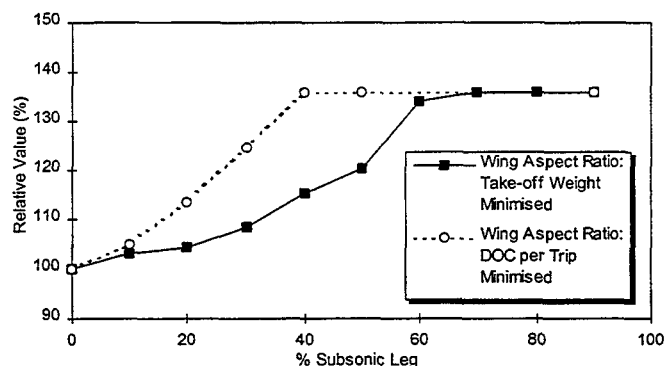


Figure 14: Variation of wing aspect ratio with subsonic leg

ASTRO was run to see how optimising for minimum direct operating costs instead of take-off weight would affect the optimum configurations. As an example, the cases from the previous study were used. For the all supersonic range case, no significant differences were found. However, as the subsonic leg proportion was increased, the aspect ratio was seen to increase more rapidly as given in figure 14. For a given case, ASTRO was now using increased aspect ratio to reduce the engine scale and fuel used whilst allowing the aircraft empty weight to increase. Take-off weight also increased but the net effect was a small reduction in operating costs.

## CONCLUSIONS

The methodology and modeling of the ASTRO SCT multivariate optimisation code have been described. The performance aspects most critical to SCT viability have been incorporated, notably airfield noise prediction which can be used to drive the design to satisfy specified noise regulations. Presented results for two short studies have demonstrated the capability to minimise aircraft take-off mass while satisfying a range of constraint types. Alternative minimisation of direct operating costs has also been briefly investigated.

A minimum take-off mass SCT aircraft meeting Stage 3 noise requirements was obtained, and its sensitivity to changes in the noise limits explored. Large mass penalties were obtained when overall aircraft redesign became necessary to accommodate reduced noise limits. A particular difficulty in accommodating reductions in the approach noise limit was identified, which could only be overcome by adopting additional noise reduction measures such as a steeper glideslope.

Subsonic cruise range requirements were shown to exert an important influence on SCT design, and particularly wing aspect ratio which varied by 35% in the presented

study. The trading of supersonic and subsonic performance with changing mission requirements is illustrated together with the corresponding variations in key design parameters. The minimisation of direct operating costs instead of take-off mass resulted most noticeably in higher aspect ratio wings, at some mass penalty, to obtain smaller engines and a lower fuel burn.

The presented studies illustrate a number of the complex interactions which occur between design variables and performance parameters, across disciplines, when seeking optimum solutions. The need for tools such as ASTRO is clearly demonstrated and development, validation and application of ASTRO will continue. Enhancements to the noise estimation method, low speed performance modeling, and mission modeling are planned to improve versatility and accuracy.

## ACKNOWLEDGMENTS

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